Cognitive planning deficit in patients with cerebellar atrophy

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Article abstract—We compared the performance of 12 patients with cerebellar atrophy (CA) and 12 normal controls matched for age and education on the Tower of Hanoi, a nine-problem task that requires cognitive planning. CA patients performed significantly worse than controls on this task despite no difference in planning and between-move pause times. A reanalysis of the data using just the subgroup of patients with pure cerebellar cortical atrophy (CCA) (N = 9) replicated the above results and also showed that CCA patients had significantly increased planning times compared with controls. Neither age, sex, education level, severity of dementia, word fluency, response time, memory, nor visuomotor procedural learning predicted CA or CCA performance. This deficit in cognitive planning suggests a functional link between the cerebellum, basal ganglia, and the frontal lobe concerning specific cognitive processes. However, the exact role of the cerebellum in cognitive planning remains undetermined.

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Patients with olivopontocerebellar atrophy (OPCA) may have a mild subcortical dementia, characterized by deficits in concept formation and learning paired-associate items, without evidence of aphasia, apraxia, or agnosia.[1] Although OPCA is a multifocal disease, the prominent damage to the *cerebellum* in OPCA patients has been linked to their impaired cognitive functioning.[2] Neuroanatomic and electrophysiologic studies indicate that the cerebellum is part of a neural system that includes the thalamus, basal ganglia, and frontal lobes.[3] Moreover, PET studies have shown a strong correlation between cerebellar and frontal cortex metabolic rates.[4] There have been numerous reports indicating that the cerebellum was involved in motor skill learning,[5-8] but until recently, there was little evidence that it might also be involved in cognitive functions.[1,2,9-11] Given the observation that the cerebellum may interact with other subcortical and cortical structures concerned with cognitive as well as motor functions,[12,13] it is likely to have some role in cognitive operations, in particular those that have been identified with prefrontal and basal ganglia functions, such as planning.

In this study, we administered to patients with cerebellar atrophy (CA) and controls a set of cognitive tasks that included a computerized version of the Tower of Hanoi, a task used to measure cognitive planning ability.[14,15] We hypothesized that patients with cerebellar lesions would have difficulty, compared with controls, in solving Tower of Hanoi problems.

Methods. *Subjects.* Twelve patients with CA were compared with 12 stratified-matched (for age and education) normal controls on a set of cognitive and motor tasks (table). The CA patients were the first 12 patients seen as part of an ongoing research program in cerebellar disorders at the National Institute of Neurological Disorders and Stroke.

Each patient was examined by an experienced clinical neurologist following the format of a standardized clinical neurologic examination. Ataxia was judged present if the examiner found incoordination of upper or lower extremity voluntary movement in the absence of weakness or sensory loss. Normal tone was evaluated on the basis of resistance to passive stretch. Autonomic dysfunction was evaluated on the basis of postural fall in blood pressure, dryness of skin, whether the patient was impotent or incontinent, and whether there was any pupillary abnormality. Each patient also underwent MRI. All scans were read by a neuroradiologist but were not quantified. The scans were judged to unambiguously demonstrate the presence or absence of CA. All CA patients demonstrated either (1) ataxia *without* CA, and additional proprioceptive sensory loss or areflexia *or* (2) ataxia *with* either sensory loss or areflexia *and* associated CA on MRI or CT. CA patients were classified into one of three diagnostic categories on the basis of this evaluation. Nine patients diagnosed with *pure cerebellar corti-*

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Table. Subject characteristics

	Normal controls	Cerebellar atrophy patients
No. of subjects	12	12
Age	52.00 (19.04)	48.37 (17.49)
Education (yrs)	15.00 (2.86)	15.25 (2.34)
Handedness	10 R/2 L	11 R/1 L
Sex	6 M/6 F	8 M/4 F
Mattis Dementia Rating Scale total score		132.75 (7.30)
Beck Depression Inventory total score		11.08 (6.28)

Values represent mean (SD).

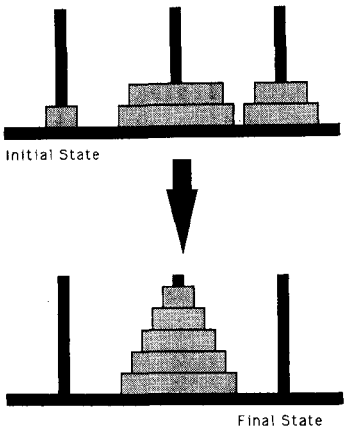

Figure. An example from the Tower of Hanoi task. Subjects see the initial state on a computer monitor. Using the number keys one, two, or three (each representing a peg), subjects move the disks to achieve the final state. Subjects can only move one disk at a time and cannot place a larger disk on top of a smaller one. Subjects get detailed instructions on how to move the disks and receive two practice problems before beginning the task.

cal atrophy (CCA) had no evidence of focal pontine atrophy on MRI or CT, had normal tone, no significant autonomic dysfunction, and no pyramidal signs. Two patients diagnosed with *OPCA* had evidence of focal pontine atrophy on MRI or CT and only mild, if any, autonomic dysfunction. The one patient diagnosed with *nonspecific multisystem atrophy* had normal tone, evidence of tremor, and significant autonomic dysfunction. Hereditary and sporadic cases were noted.

There was no significant difference between the CA patients and normal controls regarding age or education (table). There was a slight difference in the proportion of women and left-handers between the groups. The CA mean Mattis Dementia Rating Scale total score of 132 (possible range of scores is 0 to 144) suggests borderline cognitive impairment. The CA mean Beck Depression Inventory total score of 11 is within normal limits.

Tests. Tower of Hanoi.[16] The Tower of Hanoi is a cognitive planning task that requires the subject to develop and carry out a plan of action. Subjects have to transfer five disks, one at a time, from an initial state to a final state on peg 2 (see figure for an example). There are two rules that subjects have to obey: they can only move one disk at a time, and they are not allowed to place a larger disk on a smaller disk. The task is presented on a computer monitor and disks are moved using the number keys to represent the three pegs. Two practice trials are administered followed by nine different problems varying by the minimal number of moves to achieve the final state. Each problem has a different initial state and individual route to solution, minimizing reliance on a single learning rule, although a recursive strategy is embedded in the moves along the path to a solution. The difficulty of the problems that were administered to an individual subject was determined by that individual's performance on the preceding problem. The better the subject performed, the more difficult (ie, more steps to solution) the next problem became. There were, however, no significant differences in task mean problem difficulty (determined by the mean number of steps required for problem solution) between the CA patients and the normal controls. The task may induce explicit learning of the general recursive rule, implicit procedural learning, and premovement planning strategies based on the recursive rule and initial state of a problem, and requires visuomotor control to identify and press the correct keys. All subjects clearly understood the instructions as indicated by their performance on two practice problems. All subjects volunteered for this study and appeared motivated to perform the task according to the test examiner.

We compared group performance on the following measures from the Tower of Hanoi:

(1) Mean number of problems correctly solved;

(2) Mean number of illegal moves (placing a larger disk on a smaller disk);

(3) Mean planning time per problem (duration of time after the problem is shown but before the first move is made);

(4) Mean pause time per problem (time elapsed to solve a problem/number of moves per problem).

Besides the Tower of Hanoi, three other tasks were administered in order to account for any significant Tower of Hanoi performance differences on the basis of memory, visuomotor skill learning, or generation of verbal information.

Experimental Paired-Associates Test.[17] Subjects were administered words and pictures in pairs to remember. Following a brief delay, subjects were given both free- and cued-recall conditions. Then, subjects were told each item's name and had to say whether it was originally presented as a picture or a word (incidental condition).

Serial Reaction Time Task.[18] Subjects see asterisks which can appear in any one of four positions on a monitor. Whenever an asterisk appears, subjects are to press a marked computer key that falls directly underneath the position where the asterisk appeared. There were seven blocks administered. Each block includes 100 trials. Blocks one and two showed asterisks randomly presented. Blocks three to six included 10 sets of 10 repeated trials. Block seven was again a random block. Both mean reaction times and accuracy were obtained. A measure of visuomotor procedural learning can be obtained by subtracting the mean reaction time on block seven from that of block six.

Verbal fluency.[19] Subjects were required to generate as many category exemplars as possible to a category name.

Analysis procedure. For between-group comparisons, we used a one-factor analysis of variance. We also used a series of within-group simple regression analyses to determine the effects of a set of independent variables on the total number of problems solved on the Tower of Hanoi task by the CA patients. The independent variables included age, education level, motor evaluation scores, Mattis Dementia Rating Scale total score,[20] Beck Depression Inventory total score,[21] Paired-Associate learning, Verbal Fluency total score, and the Serial Reaction Time Task mean reaction time, percent correct, and procedural learning score.

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Results. CA patients solved significantly fewer Tower of Hanoi problems than normal controls ($F[1,22] = 19.288, p < 0.0002$). CA patients also had a tendency toward making illegal moves when compared with normal controls ($F[1,22] = 3.721, p < 0.06$). There were no between-group differences for planning time or mean between-move pause time per problem.

We were concerned that the significant findings may have been due to extracerebellar damage, so we reanalyzed the data using only patients with pure cerebellar lesions (CCA, N = 9). CCA patients solved significantly fewer Tower of Hanoi problems than normal controls ($F[1,19] = 13.48, p < 0.001$). CCA patients also made significantly more illegal moves than did normal controls ($F[1,19] = 4.23, p < 0.05$). Although there was no between-group difference in between-move pause time, the CCA patients took significantly longer before making their first move (increased planning time) than did controls ($F[1,19] = 4.42, p < 0.04$).

For both the CA and CCA groups, a series of simple regression analyses found no significant relationship between age, education, clinical symptoms including global motor impairment scores, dementia score, depression score, paired-associate learning, verbal fluency, or procedural learning and the number of problem solutions achieved on the Tower of Hanoi task.

Discussion. CA patients had difficulty solving problems on the Tower of Hanoi task that could not be accounted for by motor impairment, age, educational level, level of dementia, depression, visuomotor procedural learning, verbal memory, or verbal fluency. While the Tower of Hanoi has a learning component, both the use of different initial states for every problem (maximizing the use of different routes to solution) and the results of the regression analyses rule out a simple memory deficit explanation and suggest that a more likely explanation for failure on this task will be based on the specific demands of cognitive planning. Furthermore, CCA patients demonstrated a significant increase in premovement planning time (with no increase in between-move pause time) relative to the control group, suggesting that the locus of the CCA deficit on this task is memory-independent.

Convergent evidence may be able to explain CA patient impairment on this task. In addition to CA patients, there is some evidence that patients with frontal lobe lesions or frontal lobe dysfunction due to subcortical disease also fail on cognitive planning tasks similar to the Tower of Hanoi.[14,15] Such tasks differ from the Tower of Hanoi in the solution path constraints imposed by manipulating task instructions, the number and type of disks used, or by varying the height of specific pegs. Shallice[14] has shown that patients with frontal lobe lesions perform poorly on the Tower of London task. St. Cyr et al[15] found that patients with Parkinson's disease demonstrate a similar impairment on the Tower of Toronto task. The reason for these failures on tower-type tasks is unclear, although it appears that assembling a sequential series of events or actions into a coherent unit of behavior is difficult for patients with frontal lobe damage,[22] subcortical lesions,[23] and cerebellar lesions. Cognitive planning could be viewed as an analogue cognitive representation to complex motor procedures that require a series of individual movements to function as a *unitary* sequence.[24] It is possible that the cerebellar-frontal axis functions in different ways depending on the relative demands of a task. For example, motor initiation may depend on frontal cortex and motor sequences may depend on the cerebellum for the timing of movements, whereas cognitive sequences may depend on the prefrontal cortex for the initiation of plans or sequential actions and on the cerebellum for the on-line temporal integration of events that make up the cognitive plan or action.[25]

Thus, it may be that cerebellar dysfunction disrupts a neural system that includes subcortical structures and the frontal lobes. Prefrontal cortex lesions would hypothetically affect the activation of plans and actions, subcortical structures such as the thalamus and basal ganglia would allow for their automatic execution, and the cerebellum would insure their correct sequencing. In our study, cognitive sequencing problems could have affected CCA patients either when they tried to develop a planning strategy prior to actual disk movement or during their on-line performance. A breakdown in sequencing could lead to errors in selecting the correct solution path. The more novel the problem, the greater the likelihood of a planning failure since a well-established memory (which might include sequencing and decision patterns) for the plan would not exist to help compensate for on-line difficulty.

Cerebellar dysfunction alone should not affect all types of cognitive functions.[26] On the other hand, OPCA can occasionally lead to marked cognitive impairment.[27,28] In some cases, it is even suspected that subtle basal ganglia lesions may contribute to, or be the primary cause of, cognitive dysfunction in patients with cerebellar atrophy.[26] Thus, further studies contrasting the effects of cerebellar versus basal ganglia and cerebral cortex lesions on cognitive tasks requiring planning are necessary to confirm and extend our findings. When possible, these studies should be done in conjunction with brain metabolic imaging or dopaminergic receptor studies to rule out subtle prefrontal lobe, basal ganglia, or thalamic dysfunction in CCA-type patients and to correlate regional brain metabolism with different aspects of cognitive performance on planning tasks.

Addendum. Akshoomoff et al[29] and Fiez et al[30] have recently reported case studies of patients with cerebellar damage, revealing impaired cognitive functions in a number of domains. In particular,

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the patient of Fiez et al failed to learn a solution principle on the Tower of Toronto problem, a task similar to the Tower of Hanoi used in our study. Both these studies, in conjunction with the results of the experiment reported here, offer new hypotheses on the role of the cerebellum in specific *cognitive* operations.

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